

City of Greensboro — Engineering Division

300 W. Washington St., Greensboro, NC 27401

Date: March 5, 2026

Project: Battleground Business Park Phase III

Permit #: CE-2026-0045

Review Round: 1st Review

CIVIL ENGINEERING REVIEW COMMENTS — 1st Review

Reviewer: Michael Torres, PE — Civil Plan Reviewer

28 comments across 6 plan sheets

SHEET C-1 (Cover & Notes)

C1-1 [MINOR]: Benchmark referenced (NAVD88) does not match the benchmark shown on the survey. Verify and correct.

C1-2 [MINOR]: General notes do not reference the City of Greensboro Standard Specifications and Details Manual (2024 Edition). Add reference.

C1-3 [MINOR]: Vicinity map does not show the project's relationship to major roads within 1 mile. Expand map extents.

SHEET C-2 (Existing Conditions)

C2-1 [MAJOR]: Existing 12-inch sanitary sewer main shown at approximately STA 2+50 conflicts with proposed building foundation. Provide minimum 10-foot horizontal separation per Greensboro Utilities standards.

C2-2 [MINOR]: Existing utility easement (DB 4521, PG 891) not shown. Locate and show on plans.

SHEET C-3 (Grading Plan)

C3-1 [CRITICAL]: Proposed finished floor elevation (FFE) of 842.5 is only 6 inches above the 100-year floodplain elevation of 842.0. Minimum freeboard is 2 feet per Greensboro Flood Damage Prevention Ordinance Section 30-6-3.4. Raise FFE to 844.0 minimum.

C3-2 [MAJOR]: Retaining wall heights exceed 4 feet in several locations without structural engineer sealed drawings. Provide PE-sealed wall design per Section 1807 of the NC Building Code.

C3-3 [MAJOR]: Grade transitions near loading dock area exceed 10% slope for truck maneuvering area. Max 5% per AASHTO standards for commercial vehicle access.

C3-4 [MINOR]: Spot elevations not provided at all building corners. Add to clarify drainage patterns.

SHEET C-4 (Utility Plan)

C4-1 [CRITICAL]: Proposed 8-inch water main connection to existing 6-inch main on Battleground Ave will create a dead-end configuration. Per Greensboro Water Resources Policy, looped connections are required for mains serving commercial developments. Revise to connect to the 12-inch main on Industrial Parkway.

C4-2 [MAJOR]: Sanitary sewer lateral depths at the building connection points are not shown. Provide invert elevations for all laterals to verify gravity flow to the main.

C4-3 [MAJOR]: Fire hydrant spacing exceeds the maximum 500-foot distance between hydrants per NFPA and Greensboro Fire requirements. Add 2 additional hydrants.

C4-4 [MINOR]: Utility crossing detail at STA 5+20 does not show required 18-inch vertical separation between water and sewer per 15A NCAC 18C .0906.

SHEET C-5 (Stormwater)

C5-1 [CRITICAL]: Stormwater detention pond outlet structure calculations show peak discharge rate exceeding pre-development conditions for the 25-year storm. Per City requirements, post-development peak discharge must not exceed pre-development for the 2, 10, and 25-year events. Revise outlet structure sizing.

C5-2 [MAJOR]: Stormwater conveyance pipe from CB-7 to the detention pond is undersized. HGL analysis shows surcharging at the 10-year event. Upsize from 24-inch to 36-inch RCP or provide additional inlets.

C5-3 [MAJOR]: Permanent BMP (bioretention cell) drawdown time exceeds the 72-hour maximum per NCDEQ Stormwater Design Manual. Revise underdrain configuration or increase cell volume.

C5-4 [MINOR]: Riprap apron at pond outfall not shown on detail. Provide energy dissipation design per NCDOT Hydraulics Manual.

C5-5 [MINOR]: Stormwater maintenance access road to detention pond not shown. Provide 12-foot all-weather access per City standards.

SHEET C-6 (Erosion Control)

C6-1 [MAJOR]: Sediment basin sizing does not meet NPDES requirements (minimum 1,800 CF per acre of disturbed area). Site disturbance is 12.3 acres, requiring 22,140 CF minimum. Plan shows approximately 18,000 CF. Revise.

C6-2 [MINOR]: Inlet protection details not shown for all proposed catch basins during construction. Provide protection for each.

C6-3 [MINOR]: Construction entrance detail does not meet minimum 50-foot length per NCDEQ standards. Extend to 50 feet.

C6-4 [MINOR]: Silt fence alignment near the stream buffer crosses a low area that would collect water. Relocate to prevent silt fence overtopping.

Total: 28 comments (4 Critical, 10 Major, 14 Minor)

Resubmittal deadline: April 4, 2026

Michael Torres, PE

Civil Plan Reviewer, City of Greensboro